
The Missing Positional Story in LLMs: A Case Study of Shift-Invariant Attention

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Rotary positional embeddings (RoPE) are now the dominant positional mechanism
2 in modern large language models. By construction, RoPE encodes a shift-invariant
3 (SI) channel in attention: head logits can depend purely on the offset between
4 tokens rather than their absolute positions. We show that trained models do not
5 merely inherit this channel passively — they actively learn to exploit it, and
6 the exploitation is load-bearing: targeted SI interventions produce systematic
7 degradation that scales tightly with per-head SI amplitude under a converging
8 set of specificity controls, and SI structure emerges even in models trained with
9 absolute positional encodings, confirming it is a learned computation rather than
10 an architectural artifact. This SI exploitation is also functionally meaningful —
11 SI ablation preferentially disrupts offset-structured retrieval tasks across all three
12 primary models, with directional extension to naturalistic long-context settings —
13 but our results simultaneously surface a deeper question: is the SI channel currently
14 underutilized? Computations that should be position-invariant by nature, such as
15 recognizing that a mathematical equivalence holds regardless of where it appears
16 in context, are a natural domain for shift-invariant processing; yet our evidence
17 suggests current models allocate SI channels primarily toward narrow positional
18 and surface routing rather than richer invariances of this kind. Strikingly, SI
19 exploitation amplitude varies markedly even across architecturally similar models,
20 indicating the degree of use is a trainable property rather than an architectural given.
21 This opens a concrete direction: pretraining pressure toward semantic invariance
22 tasks could push models to exploit these channels for richer computation rather
23 than the narrow positional routing we observe today.

24 1 Introduction

25 RoPE gives attention a geometric route to relative-offset dependence [Vaswani et al., 2017, Su et al.,
26 2024]: under rotary transformations, query–key interactions can be expressed as functions of position
27 differences rather than absolute indices. This creates a built-in shift-invariant (SI) channel in modern
28 LLMs, but what trained models actually do with that channel remains understudied.

29 This paper studies that question as a three-model 7–8B case study. Our thesis is simple: RoPE
30 exposes an SI channel, trained models use that channel in load-bearing but heterogeneous ways, and
31 this opens a practical direction for pretraining design.

32 The primary result is the anchor contribution: SI intervention damage scales with per-head SI
33 amplitude under a converging control stack, including permutation specificity, importance-matched
34 non-SI baselines, kernel-transplant checks, and local-bias decomposition. This establishes strong
35 head-level SI specificity under intervention. The supporting functional result shows that controlled
36 offset-repetition retrieval probes are preferentially disrupted by SI-targeted ablation, with model-

conditional transfer boundaries; a two-model (Llama and Mistral) naturalistic validation extends this in the same direction on natural-text long-context retrieval. Collective ablation organization is handled appendix-first as contextual regularity (Appendix D): ranked removal rejects linear depletion, but matched non-SI controls show it is not SI-unique.

The motivating empirical surprise is heterogeneity. Across three RoPE models (Llama, Mistral, OLMo), mean SI amplitude differs by $6.5 \times (0.380, 0.266, 0.058)$, including between architecturally similar models. Matched-setup multi-seed proxy variance decomposition indicates this spread is real rather than seed noise (between-model SI variance 0.04074 vs. within-model seed variance 0.000223; ratio 182.73), but the causal drivers remain open. A breadth consolidation over 11 existing model artifacts spanning 7 families preserves the same qualitative conclusion with a much wider max/min spread, indicating this heterogeneity pattern is not limited to the primary trio.

Two forward-looking interpretive frameworks follow from this evidence. The primary lens is a *frequency-domain grounding*: RoPE inner products decompose as a Fourier sum over position offset, and the oscillatory structure of learned offset kernels predicts disruption cost empirically (Spearman $\rho=0.437$, $p=2.2 \times 10^{-13}$, Llama; Appendix B). A complementary working hypothesis is *representational commitment*: SI channels in current models appear allocated primarily to narrow positional/surface routing, with broader semantic SI usage under-realized and potentially requiring pretraining-stage pressure. A supplemental compressed-sensing-inspired lens (Appendix H) is also developed as an additional predictive framework.

2 Related Work

RoPE, positional structure, and offset-conditioned computation. Prior work shows that positional encoding choices shape what attention heads can represent [Vaswani et al., 2017, Shaw et al., 2018, Press et al., 2022, Su et al., 2024]. RoPE makes relative-offset structure especially accessible, and context-extension studies (e.g., PI and YaRN) analyze how frequency bands contribute to positional behavior [Chen et al., 2023, Peng et al., 2023]. Our focus is different: we measure per-head offset-conditioned structure and test whether that structure is load-bearing under targeted intervention, extending descriptive positional analyses [Barbero et al., 2024].

Head specialization, mechanistic intervention, and retrieval behavior. Head-level specialization and pruning work shows that some heads are functionally important while others are redundant [Clark et al., 2019, Voita et al., 2019, Michel et al., 2019, Conmy et al., 2023]. Mechanistic tools such as activation patching, causal tracing, and causal scrubbing motivate intervention-based attribution [Meng et al., 2022, Chan et al., 2022]. Induction-head and long-context retrieval literature highlights offset-structured behaviors that may rely on sparse head populations [Elhage et al., 2021, Olsson et al., 2022, Wu et al., 2024, Xiao et al., 2023]. We build on these threads with an SI-specific intervention stack, separating strong head-level specificity evidence from weaker collective and transfer-level evidence, and with a broader multi-model heterogeneity consolidation.

What prior work does not yet resolve. Existing positional and interpretability studies do not jointly answer three questions that are central here: whether SI-attributed structure is load-bearing under controlled intervention, where functional transfer boundaries lie beyond aligned toy settings, and how heterogeneous SI exploitation is across model families under a common measurement protocol. Our design targets this combined gap directly by pairing head-level intervention specificity tests with bounded functional probes and a multi-family heterogeneity consolidation.

3 Setup

3.1 SI operationalization

For each head h , we decompose pre-softmax attention logits as

$$A_h(i, j) = g_h(i - j) + r_h(i, j), \quad (1)$$

where $g_h(\Delta)$ is an offset-only component estimated nonparametrically by empirical means over all token pairs with $i - j = \Delta$. We score SI strength by

$$R_h^2 = 1 - \frac{\sum_{i,j} (A_h(i,j) - g_h(i-j))^2}{\sum_{i,j} (A_h(i,j) - \bar{A}_h)^2}. \quad (2)$$

High-SI heads are the top quartile of R_h^2 within each model. Within-model rank designation is complemented by absolute-threshold robustness checks (Appendix M): the rank-based disruption result replicates at every tested absolute threshold in the range $[0.15, 0.40]$, and per-head disruption at $T=0.20$ tracks model-level mean R^2 across all three models.

Because offset bins are frequency-imbalanced and content-position correlations can exist in natural corpora, this R^2 should be interpreted as in-distribution SI exploitation rather than corpus-free architectural SI. Pairwise weighting means common short offsets dominate the fit, while large-offset estimates are noisier due to lower support. Since R^2 is variance-normalized, cross-model contrasts are triangulated with intervention deltas in Result I; absolute nats can vary with baseline entropy, so we also report percent and control-normalized sensitivity axes (Appendix L). Completed robustness controls (Appendix L) show bounded shifts under offset reweighting and corpus-transfer scoring while preserving cross-model ordering, so this threat is reduced but not eliminated.

A key alternative interpretation is that high R^2 may partly reflect local-distance attention bias rather than richer SI computation. We therefore treat R^2 as an offset-conditioned structure score, not a standalone mechanism identifier. Local-bias decomposition controls show model-conditional support for non-trivial excess beyond simple local-decay nulls: disruption tracks R_{excess}^2 over a local-decay fit in Llama ($\rho=0.893$, $p=1.2 \times 10^{-7}$) and Mistral ($\rho=0.812$, $p=1.4 \times 10^{-5}$), but not in OLMo ($\rho=0.206$, $p=0.38$), which narrows but does not eliminate this ambiguity.

3.2 Intervention

We intervene at evaluation time by subtracting the estimated SI kernel from selected heads:

$$\tilde{A}_h(i,j) = A_h(i,j) - \hat{g}_h(i-j). \quad (3)$$

This intervention targets the offset-only pre-softmax component while leaving residual logits unchanged before softmax. Because subtraction is position-dependent, downstream attention can redistribute non-trivially through softmax renormalization. We therefore interpret measured effects as *intervention sensitivity signatures*: evidence that offset-conditioned structure is load-bearing under this perturbation class, not a full causal decomposition of redistribution pathways.

A constant-offset perturbation control provides direct evidence that position-specificity—not merely the magnitude of the kernel shift—is the load-bearing property. Applying a position-independent scalar equal to $\text{mean}(\hat{g}_h)$ per head preserves rank order and is invariant under softmax, so in theory it should not change LM loss. Empirically, this control leaves loss essentially unchanged in all three primary models: $\Delta\text{loss} = 0.00011$ nats (Llama; ratio $2913\times$ vs. true SI subtraction), 2.7×10^{-5} nats (Mistral; ratio $21968\times$), and -0.00017 nats (≈ 0 ; OLMo, where both effects are near-zero consistent with low SI amplitude). True SI subtraction causes losses of $+0.320$, $+0.592$, and $+0.015$ nats respectively. This confirms that position-specificity of the kernel—not merely the norm of the shift—drives the disruption effect across all tested models.

An entropy analysis of high-SI heads provides complementary evidence on the redistribution pathway. True SI subtraction concentrates attention (mean head-entropy change -0.887 nats in Llama, -0.719 nats in Mistral), while permuted-kernel subtraction leaves entropy approximately unchanged ($+0.031$ and -0.110 nats); in OLMo the changes are small in both arms (-0.064 vs. -0.029 nats), consistent with the lower SI amplitude. The concentration pattern in Llama and Mistral is consistent with attention-sink absorption of redistributed attention when position-specific routing is removed. Full redistribution analysis tables are in Appendix D. Importance-matched, permutation, transplant, and local-bias controls further reduce key alternatives.

3.3 Models and evidence policy

Primary models: Llama-3.1-8B [Dubey et al., 2024], Mistral-7B-v0.1 [Jiang et al., 2023], OLMo-2-7B [Groeneveld et al., 2024]. Llama and Mistral are architecturally close neighbors (both RoPE;

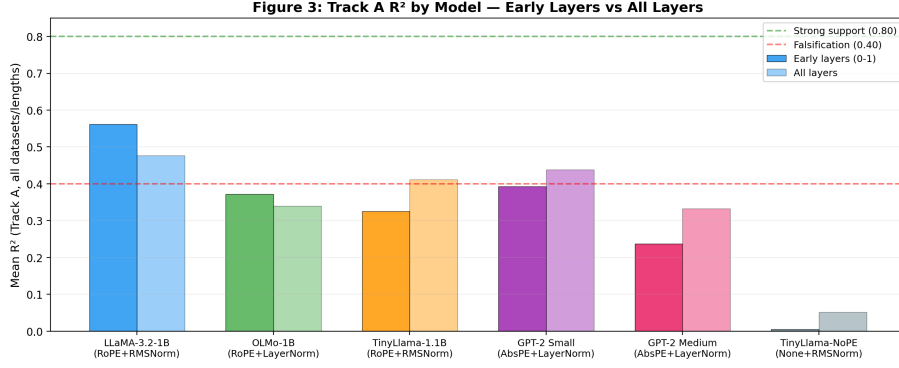


Figure 1: Setup anchor: per-head SI-strength distributions. TinyLlama-NoPE has near-zero early-layer SI, while RoPE models show broad but heterogeneous SI strength. GPT-2 anchors show non-trivial SI under absolute PE.

mean R^2 0.380 and 0.266 respectively) and provide the primary SI-specific evidence. OLMo is a more divergent weak-SI anchor (mean R^2 0.058) that provides a near-null low-amplitude reference point; cross-model claims involving OLMo describe gradient direction, not symmetric replication. Additional anchors: GPT-2-small/medium [Radford et al., 2019], TinyLlama-NoPE control, and Gemma for redundancy replication. TinyLlama-NoPE is an informative low-SI control but not a scale-matched from-scratch 7–8B NoPE training counterfactual, so we use it as a scope-bound capacity anchor only. Confirmatory results are separated from exploratory overruns; effect sizes and replication consistency are primary.

Main-paper design details (tasks, intervention units, and claim-bearing endpoints) are summarized in Appendix M (Table 20), including the 1B vs. 7–8B sourcing distinction for the NoPE anchor and TinyLlama proxy controls.

140 Main design at a glance.

- 141 • **Models:** Llama-3.1-8B, Mistral-7B-v0.1, OLMo-2-7B as the primary trio.
- 142 • **Measurement:** per-head SI R^2 from offset-only kernel fits to pre-softmax logits.
- 143 • **Primary intervention:** SI-kernel subtraction with permutation, importance-matched, transplant, and local-bias controls.
- 144 • **Primary endpoint:** head-level disruption scaling with SI amplitude under the control stack.
- 145 • **Functional endpoint:** preferential disruption on controlled offset-structured retrieval with explicit transfer-boundary checks.

148 4 Result I: RoPE Provides SI Capacity; Trained Models Exploit It 149 Heterogeneously

150 Result I is the main contribution of this paper: SI intervention damage is load-bearing and systematically stratified by per-head SI amplitude under a converging specificity-control stack.

152 RoPE-equipped models exhibit non-trivial SI structure with a large cross-model range (mean R^2 :
153 Llama 0.380, Mistral 0.266, OLMo 0.058). At 1.1B proxy scale, TinyLlama-NoPE shows early-layer
154 $R^2 < 0.01$. We treat this as weak directional capacity evidence (proxy-scale only), not a matched
155 7–8B causal positional-family contrast. Because R^2 is a ratio, we interpret this spread jointly with
156 absolute disruption deltas below rather than as a standalone measure of absolute SI magnitude. An
157 expanded breadth consolidation spanning all tested models—the three primary models plus eight
158 additional architectures across four further families (Gemma-2-9B; Qwen2.5-7B; GPT-2-small, GPT-
159 2-medium; TinyLlama-1.1B-RoPE, TinyLlama-1.1B-NoPE; Pythia-1.4B, Pythia-410M); 11 models
160 total, 7 families; Appendix L—shows mean- R^2 values ranging from 0.006 to 0.868 (max/min ratio
161 $140\times$), reinforcing that SI exploitation amplitude is broadly heterogeneous across architectures and
162 training regimes.

Head-level SI specificity hardening controls (Result I)

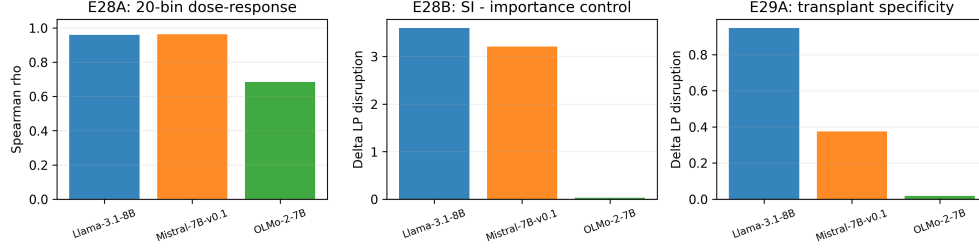


Figure 2: Head-level dose-response and specificity controls (primary finding). **Left:** 20-bin Spearman correlation between per-head R^2 and disruption cost under SI-kernel subtraction (Llama $\rho=0.959$, Mistral $\rho=0.964$, OLMo $\rho=0.684$; all one-tailed permutation $p < 0.001$). **Center:** SI-ranked intervention minus importance-matched non-SI intervention (positive in all three models, ruling out generic head-importance as the driver). **Right:** Kernel-transplant specificity — transplanting norm-matched high-SI kernels onto low-SI heads versus low-head self-subtraction (positive in all three models).

Table 1: Compact Result I hardening snapshot (main-text proof-of-stack).

Control	Main-text outcome
20-bin dose-response	Positive in all three models (Llama $\rho = 0.959$, Mistral $\rho = 0.964$, OLMo $\rho = 0.684$; all one-tailed permutation $p < 0.001$).
SI > importance-matched	SI-ranked minus non-SI importance-matched deltas are positive in all models (+3.60, +3.21, +0.0267).
Kernel-transplant specificity	Transplanted high-SI kernels on low-SI heads are more disruptive than low-head self-subtraction in all models (+0.947, +0.374, +0.0174).
Local-bias decomposition	Disruption tracks R^2_{excess} over local-decay null in 2/3 models (Llama $\rho=0.893$, $p=1.2 \times 10^{-7}$; Mistral $\rho=0.812$, $p=1.4 \times 10^{-5}$; OLMo $\rho=0.206$, $p=0.38$; non-pass).

Under SI-kernel subtraction, LM loss increases by +4.15 nats in Llama (+206%, $d = 11.6$), +3.43 in Mistral (+185%, $d = 14.6$), and +0.058 in OLMo (+2.8%, $d = 0.81$). Permutation-specificity controls (Appendix D) show true-minus-permuted effects in all three models (Llama 3.2774, Mistral 3.1926, OLMo 0.0427; all $p = 2.0 \times 10^{-4}$), indicating disruption cost specific to correct-offset structure rather than arbitrary offset subtraction.

Norm-matched perturbation controls (Appendix L) place an upper bound on this effect: in all three models, true subtraction exceeds the permuted control but is weaker than norm-matched random perturbation. We therefore frame Result I as permutation-specific disruption cost that is load-bearing under intervention, bounded above by equal-norm random perturbation.

GPT-2 anchors (absolute PE) still show measurable SI structure (Figure 1), indicating SI behavior is learned rather than a RoPE-only artifact. OLMo-only checkpoint trajectory analyses and toy-model causal probes are retained in Appendix D/M as non-central context.¹

Head-level dose-response and specificity controls (Figure 2). The control stack is consistent across models: permutation specificity is positive in all three, SI-ranked intervention exceeds importance-matched non-SI baselines (SI-minus-importance-matched deltas: +3.60, +3.21, and +0.027 nats in Llama, Mistral, and OLMo respectively), and kernel-transplant controls show that transplanting high-SI kernels onto low-SI heads is more disruptive than low-head self-subtraction. Local-bias decomposition remains model-conditional (Llama/Mistral supported, OLMo non-pass).

¹Exploratory toy-model causal probes are reported in Appendix N; they provide future-work motivation but are not scale-matched, claim-bearing evidence for Results I–III.

Table 2: SI-head ablation on a controlled offset-repetition retrieval probe. *Offset-repetition* = repeated key→value mapping (4 demonstrations); *matched-control* = independent key/value pairs, matched token count. Protocol: 200 prompts per condition per model; matched prompt length ($2 \times 4 + 1 = 9$ tokens).

Model	offset-repetition acc. Δ	matched-control acc. Δ	offset-repetition lp Δ	matched-control lp Δ	Preferential
Llama-3.1-8B	0.910	−0.005	11.121	2.612	yes
Mistral-7B-v0.1	0.715	−0.005	6.826	0.059	yes
OLMo-2-7B	0.345	−0.055	3.174	−1.401	yes

Table 3: Result III boundary snapshot (minimal main-text scope map).

Check	Outcome
Controlled probe (E12)	Directional preferential disruption in 3/3 primary models.
Strict format controls (E18)	Strict survival in 1/3 primary models.
Broader synthetic transfer (E28C)	Model-level non-pass: 0/3 pass (family-pass counts 4/7, 4/7, 2/7).
Natural-text long-context (E29B)	Directional positive in Llama/Mistral, negative in OLMo (2/3).
Boundary grid (E30A)	Same split as E29B: Llama/Mistral pass directionally, OLMo non-pass (2/3).

181 The 20-bin dose-response is positive in all three models (Llama $\rho = 0.959$, Mistral $\rho = 0.964$, OLMo
182 $\rho = 0.684$; all one-tailed permutation $p < 0.001$). Absolute-threshold robustness and multi-axis
183 comparability details are reported in Appendix M.

184 5 Result II (Context): Collective Ranked-Ablation Nonlinearity Is Robust but 185 Not SI-Unique

186 At the collective level, ranked cumulative ablation robustly rejects linear depletion across tasks and
187 models, showing that disruption accumulates nonlinearly under ordered head removal (BIC model
188 comparison: 0/18 cells prefer linear over threshold-piecewise or logistic-sigmoid fits across all three
189 primary models; full adjudication in Appendix D). However, matched non-SI controls show that this
190 collective nonlinearity is not specific to SI-ranked heads (SI-ranked 0/9 linear; matched non-SI 1/9
191 linear, delta-BIC CI spans zero). We therefore treat Result II as contextual organizational evidence
192 rather than a standalone SI-specific mechanism claim. In narrative terms, the organizational signal
193 that is SI-specific is head-level (Result I), while collective nonlinearity is a broader regularity of
194 ranked ablation behavior. Full BIC/cell-level adjudication and matched-control breakdowns are
195 reported in Appendix D and Appendix L.

196 6 Functional Sensitivity: SI Intervention Preferentially Disrupts 197 Offset-Repetition Retrieval

198 This section is a tightly scoped mechanistic sensitivity probe, not a naturalistic long-context ICL
199 benchmark. On the controlled 4-shot key→value probe used here, SI intervention is directionally more
200 damaging for offset-repetition conditions than matched controls in all 3 models by absolute accuracy
201 gaps of +0.915 (Llama), +0.720 (Mistral), and +0.400 (OLMo). In log-probability units, modelwise
202 contrasts are also positive but heterogeneous (Llama 11.121 vs 2.612; Mistral 6.826 vs 0.059; OLMo
203 3.174 vs −1.401). Semi-naturalistic long-context probe families (wiki/code passkey contexts 256–
204 512; Appendix L) show a bounded extension: pooled preferential LP gap is positive in Llama and
205 Mistral, negative in OLMo (2/3 model directional passes). A 2×2 boundary grid over context length
206 and offset-regularity (Appendix L) reproduces the same model split (Llama/Mistral directional passes,
207 OLMo non-pass). We therefore promote this as a directional preferential-degradation result under
208 intervention with explicit transfer boundaries, not as a stable scalar-ratio claim.

209 Strict format-confound controls are mixed/model-conditional (1/3 strict survival).² Broader 7-family
 210 synthetic transfer is non-passing at model level (0/3).³ A two-model natural-text long-context
 211 extension (Llama and Mistral; 60 prompts each; Appendix L) shows the same directional effect: SI-
 212 targeted ablation exceeds importance-matched non-SI ablation by +3.64 nats (Llama) and +3.12 nats
 213 (Mistral; $p=3.3\times 10^{-5}$, one-sided permutation, CI [2.21, 4.04]). The combined E28C/E29B/E30A
 214 pattern indicates a boundary: sensitivity is strongest when probes preserve explicit offset-structured
 215 retrieval under contextual distraction, and weakens under broader synthetic family diversity.

216 Result III is therefore claim-bearing as a bounded functional signal under intervention, not as a
 217 universal transfer law or a circuit-identity claim.

218 7 Robustness and Remaining Scope Limits

219 **Strongly supported.** Result I remains the strongest evidence tier: head-level SI disruption speci-
 220 ficity is positive under permutation, importance-matched, transplant, dose-response, and threshold-
 221 robustness controls. Constant-offset and entropy analyses support position-specific redistribution
 222 effects, and measurement robustness preserves cross-model ordering under reweighting and corpus
 223 transfer (Appendix D/L).

224 **Bounded or mixed.** Result III is claim-bearing but bounded: strict format controls are mixed,
 225 broader synthetic transfer is non-passing, and natural-text transfer is positive in Llama/Mistral but
 226 negative in OLMo. Result II remains contextual: collective ranked-ablation nonlinearity is robust but
 227 not SI-unique under matched non-SI controls (Appendix D).

228 **Open limitations.** Cross-model causal attribution remains open because architecture, tokenizer,
 229 corpus, normalization, and optimization vary jointly; pretraining data volume and source are the most
 230 proximal candidate drivers given the Llama/Mistral vs. OLMo split, but this is untested. Llama/Mistral
 231 provide the strongest positive signal, OLMo is a weak-SI directional anchor, and the Qwen quickcheck
 232 does not directionally corroborate. The SI metric remains in-distribution and pair-weighted, and
 233 the theoretical lenses are framework-level rather than confirmed mechanisms. Full evidence-tier
 234 definitions, promotion rules, and control artifacts are provided in Appendix L.

235 8 Discussion

236 The evidence is best read as an organizing empirical frame, not an identified causal chain: RoPE
 237 provides SI capacity, training modulates exploitation amplitude, and intervention reveals structured
 238 SI usage. Within that frame, evidence strength is stratified rather than uniform.

239 **Established empirical findings.** The central contribution is a converging head-level specificity result:
 240 across all three primary models, SI-kernel intervention is load-bearing, and per-head disruption
 241 scales with SI amplitude under permutation-specificity, importance-matched, kernel-transplant, and
 242 local-bias controls. The head-level structure is strongest in Llama/Mistral and directionally present
 243 but compressed in OLMo. A broader 11-model consolidation shows that SI amplitude heterogeneity
 244 extends well beyond the primary trio. Functional sensitivity is a supporting bound: controlled probe
 245 disruption is directionally consistent across models, with transfer scope limited to offset-structured
 246 retrieval contexts; a two-model (Llama, Mistral) naturalistic validation provides cross-architecture
 247 extension in the same direction. Cumulative ablation organization (Appendix D) is supporting context:
 248 collective nonlinearity is robust but not SI-unique.

249 **Interpretive framework.** We treat two primary lenses as complementary and provisional. The
 250 first is a *frequency-domain lens* grounded in RoPE’s structure. RoPE inner products decompose
 251 as a Fourier sum over offset: $Q_i \cdot R_{\theta}^{(i-j)} K_j = \sum_d [a_d \cos((i-j)\theta_d) + b_d \sin((i-j)\theta_d)]$. Shift-
 252 invariant heads are those where this expansion is approximately constant over offset—the offset-only
 253 component g_h dominates. Importantly, the oscillatory structure of the learned kernel g_h predicts
 254 disruption cost empirically: heads with higher DFT frequency content in their g_h kernels (richer

²Strict format controls replace offset-repetition demonstrations with random-order demonstrations matched for token count; survival requires the preferential gap to persist under this swap.

³The 7 probe families span offset-regularity and context-length combinations (full grid in Appendix L); a model passes if $\geq 5/7$ families show directional preferential disruption.

offset-specific routing patterns) show significantly larger disruption under SI-kernel subtraction (Spearman $\rho=0.437$, $p=2.2 \times 10^{-13}$, Llama; Appendix B). This connects the theoretical framework directly to the intervention evidence, and explains why removing the SI kernel is not equivalent to a generic head perturbation. The second is *representational commitment*: SI channels in current models appear stably allocated toward narrow positional/surface routing, with broader semantic SI usage under-realized. A supplemental compressed-sensing-inspired lens is developed in Appendix H. All three generate testable predictions but none is a confirmed mechanism.

Actionable future test. A concrete next step follows from both lenses: full-weight continued pretraining with explicit SI-consistency pressure on semantic invariance tasks. The prediction is that this should increase SI-linked disruption signatures on those tasks, while short-horizon fine-tuning should show weaker retargeting. Disconfirmation would be equally informative: if SI channels retarget quickly under short adaptation, or if pretraining-stage SI pressure does not shift SI-linked behavior, the commitment framing should be revised.

9 Broader Impact

Potential positive impact is methodological: the paper provides an auditable template for separating head-level specificity claims from collective-level and transfer-level claims, which can reduce over-interpretation in mechanistic interpretability studies. A second positive impact is practical hypothesis generation: if SI channels can be aligned to semantic invariances during pretraining, positional robustness and long-context retrieval may improve.

Potential negative impact is targeted degradation or steering of model behavior via structured logit-space interventions, especially if intervention recipes are over-generalized beyond their tested scope. We mitigate this by reporting non-supporting controls, separating primary from supporting evidence, and explicitly restricting claims to tested-model regularities.

10 Conclusion

This paper establishes a bounded empirical result: in three 7–8B RoPE models, SI structure is load-bearing under intervention, and head-level SI amplitude is a strong predictor of disruption cost under a converging control stack. The strongest positive evidence is concentrated in Llama and Mistral, with OLMo as a compressed directional anchor. A broader 11-model consolidation reinforces that SI amplitude heterogeneity is widespread rather than a single-pair artifact.

What remains unresolved is equally clear. We do not identify causal drivers of the $6.5\times$ cross-model SI-amplitude spread, do not provide matched-scale 7–8B RoPE-vs-NoPE causal training contrasts, and do not establish SI-unique collective organization or broad extra-family generalization. Functional probe sensitivity is model-conditional and probe-scoped; the naturalistic validation covers Llama and Mistral but not OLMo or non-RoPE architectures. Cumulative ablation organization is real but not SI-specific.

Why this matters is forward-looking: SI channels are present, load-bearing, and currently used unevenly across models. The representational-commitment and frequency-domain lenses provide a testable framework for pretraining design. The concrete prediction is that pretraining with explicit SI-consistency pressure on semantic invariance tasks should increase SI-linked functional signatures; short-horizon fine-tuning should show weaker retargeting. Validating or falsifying that prediction is the key next step, and the GPT-2 finding—that SI structure is learned even under absolute PE—suggests this pressure would find a receptive substrate even in architectures without RoPE’s geometric prior.

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357 *This document is the supplementary material accompanying the main submission. The main pa-*
358 *per reports one central contribution and one supporting functional result: (I) RoPE-provided SI*
359 *capacity is load-bearing with model-conditional exploitation amplitude, established under a con-*
360 *verging specificity-control stack; and (II) SI intervention shows directional preferential disruption*
361 *on controlled offset-structured retrieval probes with bounded transfer scope. Ranked cumulative*
362 *ablation organization is reported here in Appendix D as supporting context (collective nonlinearity is*
363 *robust but not SI-unique). The appendix provides full experimental details, additional analyses, and*
364 *secondary investigations (frequency-band interventions, fine-tuning experiments, tokenizer attribution,*
365 *frequency-domain framework, and tokenizer diagnostics/pruning) that are boundary conditions and*
366 *future-work anchors rather than load-bearing evidence for the main-paper claims. References to*
367 *equations (e.g., Eq. 1), sections (e.g., Section 2), figures, and tables prefixed with Arabic numbers*
368 *refer to the main paper. Appendix sections are labeled A, B, C, ... and are self-contained where*
369 *possible.*

370 Appendix overview

371 The appendix is organized as follows. Sections A, C, and D cover the primary evidence supporting
372 the main-paper claims. Sections B, E–I are secondary investigations providing boundary conditions
373 and future-work anchors. Section L summarizes completed claim-hardening and attribution controls
374 plus residual scale-transfer scope limits.

- 375 • **A. SI measurement** (supports central contribution): Full R^2 tables across 6 models, 3
376 datasets, 2 sequence lengths.
- 377 • **B. Frequency-band interventions** (informs frequency-domain lens): H1 supported at 7–
378 8B scale; H2 reversed (low-frequency channels serve all dependency ranges rather than
379 concentrating in SI heads), indicating that the frequency-domain lens operates at the level of
380 the learned kernel shape (g_h DFT structure) rather than strict RoPE-band segregation.
- 381 • **C. Mechanistic theory battery** (supports central contribution and functional sensitivity):
382 All 10 pre-specified theories on Llama-3.1-8B and OLMo-2-7B, including kernel ablation
383 and boundary detection.
- 384 • **D. Phase 2 adjudication** (cumulative ablation supporting context and functional sensitivity):
385 Full results for all Phase 2 sub-experiments: strict tokenizer controls, cumulative ablation
386 (non-SI-unique result), context-regime analysis, and non-RoPE anchor.
- 387 • **E. Fine-tuning stability** (boundary condition): SI structure stable under 250–300 steps;
388 SI-guided LoRA routing null.
- 389 • **F. Tokenizer attribution** (secondary): Causal perturbation evidence for the tokenizer $\rightarrow$
390 boundary $\rightarrow$ SI chain.
- 391 • **G. SI knowledge localization** (secondary): Post-hoc routing interventions; all completed
392 sub-experiments null or harmful.
- 393 • **H. Frequency-domain and predictive framework** (secondary): RoPE frequency-subspace
394 lens linking low-frequency head projections to SI amplitude; proxy coherence-gap relation
395 suggestive at small- n ($r = +0.731$, eff. $n = 6$, two-sided permutation $p = 0.141$);
396 phase-transition analogue preliminary ($r = -0.959$, eff. $n \approx 2$); expanded proxy refresh
397 mixed/non-promotional.
- 398 • **I. Tokenizer diagnostics and pruning** (secondary): BDI/BAC/KOA metrics; SI-matched
399 pruning model-conditional (Llama gain $p = 0.034$; OLMo reversal).

- 400 • **J. Statistical methodology.**
- 401 • **K. Compute resources.**
- 402 • **L. Completed claim-hardening and attribution controls:** Finalized E17/E18/E19 plus
- 403 completed E20/E21 proxy-attribution outcomes, E24/E24b/E25 Result-II/Result-I hardening
- 404 controls, E26/E27 head-level dose-response and absolute-threshold robustness, and the
- 405 E28/E29/E30/E31 hardening bundle (importance-matched, local-bias, transplant, speci-
- 406 ficity, probe-transfer, extra-model quick-anchor, breadth consolidation, coherence refresh,
- 407 functional-coupling reanalysis, and naturalistic add-on controls).
- 408 • **M. Exploratory causal probe — stage-sensitive SI pretraining (E23):** Controlled toy-
- 409 model training showing SI-consistency augmentation increases probability of offset-invariant
- 410 computation; strict stage-sensitivity hypothesis not supported; exploratory motivation only.

411 A Experiment 1: Measurement of shift-invariant structure

412 A.1 Setup

413 We measure the degree to which pre-softmax attention logits are approximated by a head-specific
 414 shift-invariant function $g(\Delta)$ across a grid of 6 models $\times$ 3 datasets $\times$ 2 sequence lengths (36
 415 combinations).

416 **Models.** Llama-3.2-1B [Dubey et al., 2024] (RoPE + RMSNorm), OLMo-1B [Groeneveld et al.,
 417 2024] (RoPE + LayerNorm), TinyLlama-1.1B [Zhang et al., 2024] (RoPE + RMSNorm), TinyLlama-
 418 NoPE-1.1B (no PE + RMSNorm, control), GPT-2 small [Radford et al., 2019] (absolute PE +
 419 LayerNorm), GPT-2 medium [Radford et al., 2019] (absolute PE + LayerNorm).

420 **Datasets.** Wikipedia (wiki40b_en_pre2019), Python code (CodeSearchNet), and synthetic random
 421 token sequences.

422 **Sequence lengths.** 256 and 1024 tokens.

423 **Measurement.** **Track A** fits $g(\Delta)$ directly to raw pre-softmax logits via least-squares regression
 424 per head, reporting R^2 as the SI metric. **Track B** uses a content-removed Gram matrix approach
 425 for cleaner positional signal estimation. Preregistered threshold: early-layer (layers 0–1) pooled
 426 $R^2 > 0.80$ for strong support; 0.40–0.80 partial; < 0.40 falsification region.

Table 4: Main-text 7–8B SI-strength anchors (reported in Result I; included here to reconcile appendix scale).

Model	Mean per-head R^2
Llama-3.1-8B	0.380
Mistral-7B-v0.1	0.266
OLMo-2-7B	0.058

Table 5: 7–8B per-head SI-strength distribution summary (same underlying head-level data used by Results I and E26).

Model	Mean R^2	Std	Q25	Median	Q75
Llama-3.1-8B	0.3803	0.1561	0.2545	0.3885	0.5040
Mistral-7B-v0.1	0.2658	0.1430	0.1556	0.2401	0.3688
OLMo-2-7B	0.0576	0.0584	0.0153	0.0457	0.0814

427 A.2 Detailed results

428 **Depth dependence.** All PE-equipped models show R^2 decay with depth. TinyLlama shows an
 429 inverted early-layer pattern (lower early, higher overall), suggesting some deeper heads develop
 430 specialized positional structure.

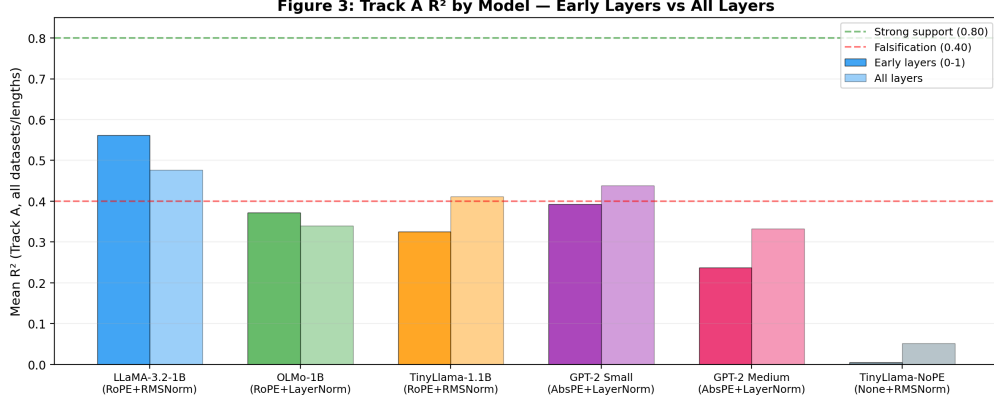


Figure 3: Per-head R^2 distributions across the five tested models (violin plots, Wikipedia/512 conditions; dashed lines mark the top-quartile cutoff defining high-SI heads). RoPE-equipped 7–8B models differ markedly in SI concentration; GPT-2 anchors (right) show intermediate R^2 without RoPE, inconsistent with a strictly RoPE-specific artifact explanation. OLMo’s narrow, low- R^2 distribution illustrates the weak-SI regime where SI-guided classification degrades. Referenced in main text Section 3.

Table 6: Track A early-layer R^2 by model, dataset, and sequence length (selected rows).

Model	Dataset	Len	Early R^2	Overall R^2
Llama-3.2-1B	synthetic	256	0.625	0.544
Llama-3.2-1B	synthetic	1024	0.621	0.596
Llama-3.2-1B	wiki	256	0.517	0.424
Llama-3.2-1B	wiki	1024	0.523	0.414
OLMo-1B	synthetic	256	0.453	0.388
OLMo-1B	wiki	1024	0.346	0.300
TinyLlama-NoPE	synthetic	256	0.008	0.023
TinyLlama-NoPE	wiki	1024	0.002	0.074

Dataset sensitivity. Synthetic sequences yield the highest R^2 (~ 0.35 averaged across models), followed by Wikipedia (~ 0.30) and code (~ 0.30).

Track A vs Track B. Track B centered diagnostics confirmed parity with CPU baselines at machine epsilon across a 42-job centering granularity sweep.

B Experiment 2: Causal frequency interventions

B.1 Setup

Hypotheses. H1: High-frequency ablation hurts short-range more. H2: Low-frequency ablation hurts long-range more.

Intervention. Partition D RoPE frequency pairs into high/low bands. Strong ablation removes 60% of targeted-band channels. Continuous attenuation at 25%, 50%, 75% severity rules out hard-zero artifacts. Norm-preserving rescaling (drift ≤ 0.05).

Task suite. Local copy-offset (short), local key-match (short), delayed copy (long), long-range retrieval (long).

Models. 1B: Llama-3.2-1B [Dubey et al., 2024], OLMo-1B [Groeneveld et al., 2024], TinyLlama-1.1B [Zhang et al., 2024]. 7–8B: Llama-3.1-8B [Dubey et al., 2024], OLMo-2-7B [Groeneveld et al., 2024], Mistral-7B [Jiang et al., 2023].

B.2 Detailed results

Table 7: Complete Experiment 2 results across all analysis branches.

Analysis	H1 effect	H1 status	H2 effect	H2 status
1B confirmatory (3 models)	+0.047	near-miss	−0.155	reversed
1B feasibility (Llama + OLMo)	+0.097	pass	−0.166	reversed
7–8B Phase 2A (3 models)	+0.112	pass	−0.245	reversed
Option D ablation (7–8B)	+0.101	pass	−0.205	reversed
Option D attenuation 75%	+0.094	pass	−0.195	reversed
Option D attenuation 50%	+0.082	pass	−0.182	reversed
Option D attenuation 25%	+0.074	pass	−0.132	reversed

H2 interpretation. The reversal is not an artifact of hard-zero ablation (attenuation replicates), span overlap (fixed in Option D), or coarse averaging (pair-level shows same direction). Low-frequency channels serve all dependency ranges.

C Experiment 3: Mechanistic theory battery

C.1 Setup

Ten competing theories tested on Llama-3.1-8B [Dubey et al., 2024] and OLMo-2-7B [Groeneveld et al., 2024]. High-SI heads defined as top quartile by per-head R^2 .

Table 8: Full Experiment 3 theory battery results. Theories T2, T4, and T6 were not executed as part of the core Experiment 3 battery in this program; they appear as reserved entries in the theory numbering scheme.[†]

Theory	Hypothesis	Llama	OLMo	Key metric
T1	SI $\leftrightarrow$ task perf.	Not supp.	Not supp.	$\Delta\text{drop} = -0.015, p > 0.97$
T3	Cross-term corr.	Descriptive	Descriptive	$\rho: -0.11 / +0.05$
T5	Subword continuation	Not supp.	Not supp.	Interaction reversed
T5b	Boundary detection	Supported	Supported	All criteria met
T7	Induction feeders	Supported	Supported	High-SI feeds induction
T7b	Activation patching	Not supp.	Supported	$\rho = 0.06 \text{ (NS)} / 0.21$
T8	Kernel ablation	Supported	Supported	Significant loss increase
T9	Freq-band feeders	Descriptive	Descriptive	$\rho_{\text{HF}} = 0.44 / 0.21$
T10	Feeder specificity	Not supp.	Not supp.	Not induction-specific

[†]T2, T4, and T6 are reserved placeholder slots in the preregistered theory numbering schema: T2 for causal training-dynamics tests, T4 for cross-layer propagation tests, and T6 for multi-model circuit-identity tests. These slots were not executed in the core battery for this submission window. All executed theories appear in this table; no executed theory is excluded. The complete preregistration schema, including all reserved and deferred slots, is available in the supplementary preregistration artifacts (Appendix J).

Note on T1 vs. T8. T1 failure (no per-head SI–task correlation; $p > 0.97$) is consistent with T8 support (significant collective kernel ablation): when SI contributions are small and redundant across heads, per-head SI–performance correlation is expected to be near zero even as collective ablation cost is large. T1 non-significance is therefore structurally predicted by Result III (distributed redundancy), not a contradiction of it.

T5b details. Boundary detection criteria: (A) boundary attention $>$ chance, (B) high-SI $>$ low-SI boundary attention, (C) effect survives content controls. All three criteria met for Llama; A and C met for OLMo with B marginal.

T8 details. Positional kernel $\hat{g}(\Delta)$ estimated per head via least-squares, then subtracted from logits. Loss increase significant in both models.

T7/T10 interpretation boundary. T7 support means high-SI heads contribute to pathways that include induction heads. T10 non-support means this feeder relationship is not induction-specific: high-SI heads also feed non-induction targets. Accordingly, we use T7 as compatibility evidence for ICL-relevant routing, but do not claim an induction-unique SI circuit identity.

D Experiment 3 Phase 2: Adjudication layer

Eleven sub-experiments targeting unresolved mechanisms from the core battery.

Table 9: Phase 2 adjudication: full results across 11 sub-experiments.

Exp	Target	Key outcome
3P2-A	Positional broadcast	Model-conditional: Llama specialized, OLMo uniform
3P2-B	Boundary triviality	Non-trivial baseline signal; strict status model-conditional (Mistral clean, OLMo directional-null caveat, Llama gate-ambiguous)
3P2-B ext	Tokenizer feature audit	Llama expanded feature ablations block/reverse boundary signal, consistent with tokenizer-mediated routing
3P2-C.1	Redundancy model	Linear rejected in all 3-way cells (0/18 linear; 12/18 threshold, 6/18 logistic)
3P2-C.2	Superlinearity test	No superlinear support
3P2-C.3	Low-SI backup	Directional support for distributed backup
3P2-D	Architecture tiebreaker	Weak unique-SI after prev-token control (Mistral)
3P2-E	T5 vs T5b reconciliation	Boundary-primary resolution
3P2-F	Proxy R^2 confound	Retained ($\Delta R^2 = 0.003$); underpowered
3P2-G	Dose-response saturation	Mixed: OLMo early slope, Llama null
3P2-H	Source/target spec	Mixed: Llama recovered specificity, OLMo null
3P2-I	Tokenizer invariance	OLMo strict-pass (0.75); Llama strict-ambiguous (neither pass nor blocked)
3P2-J	Conditional specialization	Interaction significant in both models; proxy-to-task sign transfer not yet stable
3P2-K	Non-RoPE anchor	SI detected in GPT-2-medium and GPT-2-small; strict non-trivial boundary signal in both

3P2-C.1 details. All three models (Llama, OLMo, Mistral) give zero BIC votes to linear degradation in a 3-way comparison (threshold-piecewise, logistic-sigmoid, linear). Per-model votes: 4/2/0 (threshold/logistic/linear), identical across all three; totals are 12/18 threshold, 6/18 logistic, 0/18 linear. Both non-linear models are consistent with the distributed-redundancy interpretation: threshold-piecewise captures abrupt collapse; logistic-sigmoid captures smooth accelerating collapse converging to the same collective-depletion structure. This is the primary evidence for the distributed redundancy claim in the main paper. The threshold finding is also directionally consistent with the phase-transition prediction in Exp. 7B (see Appendix H).

E5 fourth-family protocol details (Gemma). E5 executes on google/gemma-2-9b with the same cumulative-ablation fraction grid $\{0, 0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.50\}$ and 5 seeds. Per-head SI strength uses $n = 500$ random-sequence evaluations (length 128; max offset 64). Kernel-ablation LM loss uses 40 random sequences (length 128). Cumulative-ablation curves use 40 key-match and 40 long-range retrieval prompts per seed, plus LM-loss evaluations on up to 20 random sequences per fraction/seed. Observed BIC votes are threshold in all 3 task cells (0/3 linear).

D.1 Reinforcement experiments (R1–R5 + follow-ups)

Evidence timestamp and provenance. Appendix reinforcement evidence includes artifacts through April 28, 2026. This includes the completed R3 core replication report and the follow-up reinforcement controls NEW-R12, NEW-R14, NEW-R15, R2B, and R5B.

495 **Full artifact paths.**

- 496 • R1: results/reinforce_exp/exp_r1_multidomain_gate/multidomain_gate_
497 summary_v1.json
- 498 • R2: results/reinforce_exp/exp_r2_dependence_reanalysis/claim_
499 stability_table.json
- 500 • R3: results/reinforce_exp/exp_r3_core_replication/mistral-7b-v0.1/
- 501 • R4: results/reinforce_exp/exp_r4_pe_scheme_contrast/pe_scheme_
502 comparison.json
- 503 • R5: results/reinforce_exp/exp_r5_task_grounded/interaction_
504 comparison_proxy_vs_task.json
- 505 • R2B: results/reinforce_exp/exp_r2b_olmo_boundary_power/olmo-2-7b/
506 claim_impact.json
- 507 • R5B: results/reinforce_exp/exp_r5b_regime_alignment/interaction_
508 transfer_report.json
- 509 • NEW-R12 (Llama): results/reinforce_exp/exp_new_r12_ordering_control/
510 llama-3.1-8b/summary.json
- 511 • NEW-R12 (OLMo): results/reinforce_exp/exp_new_r12_ordering_control/
512 olmo-2-7b/summary.json
- 513 • NEW-R12 (Mistral): results/reinforce_exp/exp_new_r12_ordering_control/
514 mistral-7b-v0.1/summary.json
- 515 • NEW-R14 (Llama): results/reinforce_exp/runs/neurips_ext_opt_20260427_
516 134753/exp_new_r14_kernel_permutation/llama-3.1-8b/specificity_test.
517 json
- 518 • NEW-R14 (Mistral): results/reinforce_exp/runs/neurips_ext_opt_20260427_
519 134753/exp_new_r14_kernel_permutation/mistral-7b-v0.1/specificity_
520 test.json
- 521 • NEW-R14 (OLMo): results/reinforce_exp/runs/neurips_ext_opt_20260427_
522 134753/exp_new_r14_kernel_permutation/olmo-2-7b/specificity_test.
523 json
- 524 • NEW-R15 (Llama): results/reinforce_exp/exp_new_r15_head_stability/
525 llama-3.1-8b/stability_summary.json
- 526 • NEW-R15 (OLMo): results/reinforce_exp/exp_new_r15_head_stability/
527 olmo-2-7b/stability_summary.json

528 **R1 prereg decision rule (explicit).** The multidomain strict adjudication uses two preregistered
529 aggregate gates recorded in the R1 artifact under decision_rule. Let assessment_domain be
530 the per-domain strict status from domain-level multiseed summaries. Aggregate stable_blocked
531 requires:

$$\#\{\text{stable_blocked domains}\} \geq 2, \quad (4)$$

$$p_{\text{meta}} < 0.01, \quad (5)$$

$$\Delta_{\text{pooled mean}} \geq \delta_{\text{min}}. \quad (6)$$

532 Aggregate stable_clean requires:

$$\#\{\text{stable_clean domains}\} \geq 2, \quad (7)$$

$$\#\{\text{stable_blocked domains}\} = 0, \quad (8)$$

$$p_{\text{meta}} > 0.2, \quad (9)$$

$$\Delta_{\text{pooled mean}} \leq 0. \quad (10)$$

533 If neither gate passes, the preregistered outcome is ambiguous (the observed R1 outcome).

534 **R5 prereg transfer rule (explicit).** R5 uses a two-gate confirmatory rule for regime-transfer
535 interpretation:

- 536 1. Interaction gate: SI-group $\times$ regime interaction must be significant in each model after
537 family correction (`interaction_model.p_value_holm` < 0.05).
- 538 2. Transfer gate: proxy-to-task sign concordance must satisfy `n_matches` / `n_compared`
539 $\geq 3/4$ per model (`proxy_vs_task_sign_match` fields).

540 In the current R5 artifacts, the interaction gate passes in both models while the transfer gate fails
541 (Llama 1/4, OLMo 2/4), yielding the proxy-specific semantic interpretation used in main text.

542 **R2 dependence-aware sensitivity caveat.** EXP-R2 preserves the directional signal for 7 of
543 8 audited claim cells, strengthening the overall causal picture under sequence-cluster sen-
544 sitivity checks. However, OLMo’s strict boundary non-triviality check in the R2 frame
545 (`B_boundary_nontriviality`) is direction-unsupported. Accordingly, OLMo boundary claims in
546 main text are presented as supported with caveat rather than fully unconstrained support.

547 **R5 proxy-to-task discrepancy (required scope downgrade).** EXP-R5 confirms SI-group $\times$ regime
548 interaction remains statistically significant for both primary models in task-grounded settings. But
549 the preregistered transfer criterion for regime-direction stability fails: proxy-to-task sign concordance
550 is 1/4 for Llama and 2/4 for OLMo, below the 3/4 acceptance threshold. This does not remove the
551 interaction claim; it narrows interpretation to interaction-level specialization with proxy-specific
552 regime semantics pending replication.

553 **R5B reinforcement update (required scope retained).** R5B repeats transfer analysis with tighter
554 regime mapping and increased paired support. Interaction remains significant in both models,
555 but transfer still fails: Llama 1/2 and OLMo 1/2 sign matches (criterion remains $\geq 3/4$). This
556 independently confirms that the interaction-level claim is robust while regime-direction semantics
557 remain proxy-specific.

558 **R3 completion outcome (resolved).** EXP-R3 now provides `core_replication_report.json`
559 with a positive headline: kernel load-bearing is replicated, strict-B prefix artifact is negative, and
560 C1 threshold support replicates in Mistral. Accordingly, this work now supports “three-model core
561 battery replication” for T8/B/C1-level claims, while maintaining conservative attribution language
562 because model-family differences are not isolated causally. This R3 C1-threshold signal should be
563 read as endpoint-specific and does not override the separate `reinforce_exp2` C1 two-carrier-class
564 mixed/null outcome.

565 **NEW-R12 random-order control.** Threshold preference remains robust un-
566 der random head-removal orderings: Llama 7/10, Mistral 9/10, and OLMo
567 10/10 random orderings (`aggregate_threshold_majority=26/30`), supporting
568 `robustness_assessment=strong_replication`.

569 **NEW-R14 kernel-specificity control.** In all three primary models, true SI-kernel perturbation
570 effects exceed offset-permuted controls: Llama mean true-minus-permuted difference = 3.2774 (95%
571 CI [3.2138, 3.3415]), Mistral = 3.1926 (95% CI [3.1416, 3.2486]), and OLMo = 0.0427 (95% CI
572 [0.0329, 0.0541]), with one-sided bootstrap $p_{\text{one}} = 2.0 \times 10^{-4}$ in each model. This directly supports
573 perturbation specificity for the T8 mechanism channel across the primary set.

574 **NEW-R15 head-identity stability control.** High-SI head identity is partially stable across domains
575 in both models (Llama three-way Jaccard = 0.594, OLMo = 0.427; both above random expectation),
576 with strong pairwise SI-rank correlations. Because high-confidence stability criteria are not met
577 and the code-domain sample is small, these results strengthen but do not fully lock head-identity
578 invariance claims.

579 **E12 protocol details (reproducibility clarification).** E12 uses 3 primary models, 200
580 prompts/condition/model, and 2 conditions: `icl` and `non_icl`. ICL prompts are 4-shot key $\rightarrow$ value
581 demonstrations followed by a query key; non-ICL prompts match token count but break mapping

consistency via independent key/value pairs and a novel query key. Evaluation reports both accuracy and mean target-token log-probability under intact vs. high-SI-head ablated models. Denominator guards are preregistered: $\text{abs}(\text{non_icl_acc_loss}) \geq 0.01$ and $\text{abs}(\text{non_icl_lp_loss}) \geq 0.05$. Model-level support is assigned by preferential-degradation votes over valid metrics, so directional vote outcomes are primary and ratio magnitudes are secondary.

D.2 reinforce_exp2 Path A/B consolidation (as of April 24, 2026)

Evidence window and scope. This subsection summarizes reinforce_exp2 Path A/B artifacts with evidence window closed on April 24, 2026 (America/New_York). Full Path C analysis remains excluded from this revision; we include only a status-only note for C1/C2 below.

Path C status-only note (as of April 25, 2026). C2 has a strict endpoint-null outcome ($\text{C2_supported}=\text{false}$; C2.1 and C2.2 fail strict gates) and is kept mixed/non-promotional at claim tier. C1 is now completed and returns $\text{C1_two_carrier_class_supported}=\text{false}$ (criteria 1–4 fail; criterion 5 disjointness passes), so C-path evidence remains mixed and is not promoted to core claim support in this manuscript revision.

B1 mixed outcome (criteria split). B1 passes reliability/structure requirements (criteria 1/2/6) but fails depth-controlled enrichment and secondary concordance gates (criteria 3/4/5). The intended-mode Mistral shard-wise reliability rerun keeps this status unchanged ($\text{n_valid_layers}=0$ with the available 200-sequence calibration split). This supports using taxonomy outputs as structured exploratory context, not as standalone confirmatory mechanistic proof.

B2a mixed outcome (directional heterogeneity under Holm correction). B2a shows strong model-specific signals in opposite channels (e.g., Llama absolute alignment Holm-corrected 1.75×10^{-16} ; Mistral signed alignment Holm-corrected 6.60×10^{-15}), while OLMo is null under corrected thresholds (absolute/signed Holm-corrected 0.823). Because preregistered cross-criterion consistency is not met, B2a remains mixed.

B3 inconclusive outcome (insufficient clusters). B3 returns B3_inconclusive with reason $\text{insufficient_clusters}$; model-level mechanism verdicts for eligible models are “neither” and Holm family p-values are 1.0. Hence redundancy-vs-group disambiguation is not promoted beyond conservative threshold-capacity wording.

B4 mixed outcome (cross-model asymmetry). B4 is asymmetric at model level: Llama passes retention ($\text{retention_ok}=\text{true}$), while OLMo and Mistral do not. Because preregistered cross-model retention criteria fail at family level, B4 remains mixed and non-load-bearing for central claim promotion.

E Experiment 4: SI-guided fine-tuning

E.1 Setup

Tests whether SI structure can be strategically leveraged during fine-tuning for mathematical reasoning. Three sub-experiments in gated progression.

Models. Llama-3.1-8B [Dubey et al., 2024] and OLMo-2-7B [Groeneveld et al., 2024] for both 4A and 4C.

Training. QLoRA with attention/MLP adapter injection, 250–300 steps, batch size 1, learning rate 2×10^{-4} . Math training mix: digit arithmetic, counting chains, modular arithmetic, sequence continuation (120 examples/task) + 20% wiki/instruction control.

Evaluation. Math accuracy is measured as 4-shot accuracy on a 200-example subset of the MATH benchmark [Hendrycks et al., 2021], drawn from the *Number Theory* and *Counting & Probability* categories. These categories were selected for their alignment with the training mix (arithmetic and combinatorial reasoning). Results may not generalize to harder MATH categories (Algebra Level

627 4–5, Geometry, Precalculus). The observed 59–66% accuracy range reflects the relatively accessible
628 nature of this subset.

629 **SI head definition.** Top quartile by per-head R^2 from Experiment 3 Phase 2.

630 E.2 Results

631 **4C: SI trajectory under fine-tuning (Llama-3.1-8B and OLMo-2-7B).** Snapshots at steps 0,
632 50, 100, 150, 200, 250, 300 show minimal drift in both models: Llama mean $\Delta R^2 = +0.0019$,
633 boundary- d change = -0.016 ; OLMo mean $\Delta R^2 = -0.0010$, boundary- d change = -0.012 .
634 Head-rank ordering is preserved throughout. SI structure is stable in the short-to-medium fine-tuning
635 regime; behavior under extended fine-tuning (thousands of steps) remains untested.

636 **4A: SI-aware LoRA comparison (cross-model).** 30 runs (5 conditions $\times$ 3 seeds $\times$ 2 models):
637 All conditions are null in both models (Llama means: 0.864–0.887; OLMo means: 0.630–0.647). 4B
638 (SI-loss-augmented training) deferred pending 4A signal.

639 F Experiment 5: Tokenizer effect audit

640 F.1 Setup

641 Four sub-experiments auditing tokenizer dependence and boundary-linked SI behavior. These
642 analyses are model-conditional diagnostics, not a full causal decomposition of cross-model SI-
643 strength heterogeneity.

644 **5A.** 6 models across 4 tokenizer families (GPT-2, Llama, OLMo, Pythia).

645 **5B.** Llama-2-7B [Touvron et al., 2023] vs. Llama-3.1-8B [Dubey et al., 2024]: same architecture
646 family, different tokenizer (32k BPE vs. 128k BPE). Note that these models also differ in training
647 data, scale, and RLHF; 5B provides correlational evidence, with strict causal attribution depending
648 on 5C.

649 **5C.** 12 perturbation cells $\times$ 2 models: fake boundaries, character decomposition, morpheme breaks,
650 boundary merges, random resegmentation.

651 **5D.** 5 domains: wiki (baseline), code, dialogue, technical, adversarial BPE.

652 F.2 Results

653 **5B: Head identity shuffles with tokenizer.** Top-quartile SI head overlap between Llama-2 and
654 Llama-3.1: Jaccard $J = 0.177$ (17.7% shared). Rank correlation $\rho = 0.048$ (non-significant).
655 Changing the tokenizer while holding architecture family constant substantially reassigns which
656 heads develop SI structure; note the confound with other model differences (see Setup above).

657 **5C: Causal perturbation results.** Tests run on 12 perturbation cells $\times$ 2 models (Llama-3.1-8B
658 and OLMo-2-7B), evaluating mean SI head attention weight at boundary vs. continuation positions
659 across $n \approx 512 \times 24 = 12,288$ position-pairs per model (24 sequences $\times$ 512 positions). The large
660 n explains the extreme p -values; the effect sizes ($\Delta \approx +0.017$) are small in absolute terms. No
661 correction for the 12-cell family is reported.

662 These perturbations support tokenizer mediation of boundary-linked SI behavior in the tested model
663 configurations, but do not identify tokenizer vocabulary overlap as the driver of cross-model SI
664 amplitude differences. Note that the extraordinarily small p -values reflect the large n , not large effect
665 sizes; the practical effect ($\Delta \approx 0.017$) should be interpreted in context.

666 **5D: Domain robustness.** Llama: dialogue $+0.057 R^2$; technical $-0.040 R^2$. OLMo: dialogue
667 $+0.027$; technical -0.004 (more stable). SI is domain-general infrastructure with ± 0.06 variation
668 across very different text types including adversarial BPE sequences.

669 **5A: Cross-family.** SI present in all 4 tokenizer families. Pythia anomaly: NaN boundary depen-
670 dence indices despite 80–99% classifier accuracy; 4/6 models have finite boundary measurements.

671 **G Experiment 6: SI-guided knowledge localization**

672 **G.1 Setup**

673 Two completed intervention strategies test whether localizing math computation to SI channels
674 improves reasoning performance in the finalized evidence set:

- 675 1. 6A: Gradient routing (math gradients exclusively through SI head parameters)
- 676 2. 6B: Anti-localization loss (penalize attention localization in non-SI heads)

677 Exploratory placeholders 6C–6E (two-phase SI distillation, contrastive channel assignment, SI-
678 optimized tokenizer formatting) are excluded from this submission’s finalized claim-bearing evidence
679 set.

680 Models: Llama-3.1-8B [Dubey et al., 2024] and TinyLlama-1.1B [Zhang et al., 2024] (6A, 6B, 6C);
681 OLMo-2-7B [Groeneveld et al., 2024] (6A). 3 seeds per model per condition. Evaluation: math
682 accuracy (4-shot), wiki perplexity.

683 **G.2 Results**

684 **6A: Gradient routing.** At 8B scale, math accuracy is significantly harmed while language modeling
685 improves. The scale asymmetry reflects that larger models have optimized SI allocations that routing
686 disrupts.

687 **6B: Anti-localization.** TinyLlama: math $\Delta = -0.5$ pp ($p = 0.65$, null). Llama-3.1-8B seed 0:
688 math $\Delta = -4.4$ pp (directional, partial data).

689 **Submission scope note.** Main-paper claims use only completed 6A/6B outcomes. Within com-
690 pleted sub-experiments, no positive math-accuracy gain is observed.

691 **Combined interpretation.** Among the completed sub-experiments (6A: gradient routing; 6B:
692 anti-localization), both are null or harmful. This supports a conservative, scope-limited conclusion:
693 “every completed Experiment 6 sub-experiment is null or harmful in the tested settings.” Combined
694 with Exp. 4’s null LoRA routing and immutability results across two model families, the available
695 evidence suggests SI is difficult to productively redirect with post-hoc routing constraints in the
696 evaluated regime.

697 **H Experiment 7: CS-inspired framework validation**

698 **H.1 Setup**

699 **7A: Welch-gap proxy relation (audited).** For each model and sequence length, we compute
700 analytic proxy coherence descriptors from PE metadata (PE type, head dimension, and RoPE θ
701 schedule), then join these descriptors with Track-A mean $\bar{R}^2$. Because coherence descriptors are
702 repeated across dataset rows, the primary analysis is model-level ($n = 6$ unique models). The 36-row
703 table (6 models $\times$ 2 sequence lengths $\times$ 3 datasets) is retained only as exploratory pseudo-replicated
704 context.

705 **7B: Phase-transition analogue.** Models: Llama-3.1-8B [Dubey et al., 2024] and OLMo-2-
706 7B [Groeneveld et al., 2024]. Sequence lengths: 256, 512, 1024. Protocol: progressively ablate
707 SI heads ranked by R^2 , evaluate at each fraction (0, 5, 10, 15, 20, 25, 35, 50, 75% ablated), 3
708 seeds $\times$ 128 sequences per seed. Compute observed critical point as inflection of degradation curve.
709 Compare to $m^* = C \cdot s_{\text{eff}} \cdot \log(N/s_{\text{eff}})$ where $s_{\text{eff}} = N \cdot \bar{b}$. In the corrected canonical run, C uses
710 single-condition calibration on Llama-3.1-8B at sequence length 256 ($C = 7.057$), with 5 remaining
711 conditions reported out-of-sample. A global-fit reference ($C = 7.834$) is reported separately.

H.2 Results

7A: Coherence-gap proxy relation is mixed at model level. This analysis is now explicitly proxy/model-level and non-promotional. The model-level relation is directionally positive but not conventionally significant, and the 36-row correlation is exploratory because coherence features repeat by model/length. We therefore retain 7A as a framework-building signal rather than confirmatory explanatory evidence.

E31B refresh note (expanded proxy panel). An expanded coherence refresh over an 11-model breadth panel (Appendix L, E31B) is mixed: Pearson $r = 0.577$ (two-sided $p = 0.063$), Spearman $\rho = 0.500$ ($p = 0.117$), and permutation two-sided $p = 0.070$. Direction is consistent with 7A, but significance is non-decisive at this panel size. We therefore retain the coherence-gap lens as framework-level and non-promotional for mechanism claims at current scale.

7B: Phase-transition critical point. Using canonical single-condition calibration ($C = 7.057$ on Llama-3.1-8B, length 256), Pearson $r = -0.959$ ($p = 0.0025$) and Spearman $\rho = -0.878$ ($p = 0.021$) between predicted and observed thresholds across 6 conditions. Out-of-sample (excluding the calibrated condition), Pearson remains strong ($r = -0.986$, $p = 0.0019$) while Spearman is directional but marginal ($\rho = -0.866$, $p = 0.058$, $n = 5$). BIC model comparison remains threshold-piecewise preferred in 6/6 cases (consistent with 3P2-C.1 unanimous preference). This remains preliminary evidence ($n = 6$ conditions, effective $n \approx 2$ independent model families due to sequence-length correlation within models) requiring replication across additional distinct model families.

I Experiment 8: Tokenizer analysis via SI structure

I.1 Setup

8A: Boundary discrimination metrics. Models: GPT-2-small [Radford et al., 2019], TinyLlama-1.1B [Zhang et al., 2024], Llama-3.1-8B [Dubey et al., 2024], OLMo-2-7B [Groeneveld et al., 2024]. 24 sequences of length 512 (Wikipedia). Boundary definition: position t is a word boundary iff $\text{word_id}[t] \neq \text{word_id}[t - 1]$ (using tokenizer fast API word IDs). Three metrics per model:

- **BDI**: mean $A_h(t, t-1)$ at boundary minus at continuation positions.
- **BAC**: $\text{BDI}_{\text{SI}} / \text{BDI}_{\text{non-SI}}$.
- **KOA**: fraction of $\{d_{b_j} - d_{b_{j-1}}\}$ that are multiples of head’s dominant kernel offset $\Delta^* = \arg \max_{\Delta} \hat{g}(\Delta)$.

Preregistered hypotheses: H1: $\text{BDI}_{\text{SI}} > 0$; H2: $\text{BDI}_{\text{SI}} > \text{BDI}_{\text{non-SI}}$; H3: $\text{BAC} > 1$; H4: $\text{KOA} > \text{chance}$.

8B: Tokenizer-aware pruning. Models: Llama-3.1-8B [Dubey et al., 2024] and OLMo-2-7B [Groeneveld et al., 2024]. Five mask types $\times$ 9 ablation fractions (0, 5, 10, 15, 20, 25, 35, 50, 75% ablated). Mask types: (1) `si_matched`: remove lowest- R^2 heads per model’s own tokenizer; (2) `si_mismatched`: use cross-model SI ranking (Llama mask applied to OLMo, vice versa); (3) `entropy`: remove lowest-entropy heads; (4) `activation`: remove lowest-activation-norm heads; (5) `random`: matched-cardinality random removal. Evaluation: 4-shot math accuracy (MATH subset) and wiki perplexity. Area under degradation curve (AUDC) and paired t -tests across fraction levels.

I.2 Results: 8A

Primary BDI/BAC/KOA runs are complete on small models (GPT-2-small and TinyLlama-1.1B). 7–8B BDI/BAC/KOA results are treated as non-primary in this submission snapshot and are not used for preregistered 8A claim adjudication. An exploratory all-model SIBAS variant was run separately and is not used for the preregistered BDI/BAC/KOA claims reported here.

H3 ($\text{BAC} > 1$) is not supported as preregistered: BAC is negative because non-SI BDI is slightly negative. The unexpected negative BAC is a secondary observation requiring follow-up; it is not a stronger version of the preregistered hypothesis.

H4 (KOA>chance) requires validation against a proper null distribution (random boundary spacings or low-SI heads). KOA values near 1.0 for both models require this calibration before H4 is interpretable. BDI $\times R^2$ Spearman $\rho = 0.407$ (GPT-2, $p < 0.001$) and 0.329 (TinyLlama, $p < 0.001$) confirm boundary discrimination as continuous in SI degree.

I.3 Results: 8B

The pre-specified primary comparison is SI-matched vs. SI-mismatched ($\alpha = 0.05$, no correction): $p = 0.034$ on Llama-3.1-8B. Exploratory comparisons: matched vs. entropy $p = 0.003$; matched vs. activation $p = 0.017$; matched vs. random $p = 0.337$ (ns). The paired t -test treats ablation fractions as independent paired observations, which may not fully account for serial correlation in the degradation trajectory; AUDC is the primary summary statistic and t -tests are secondary.

OLMo-2-7B shows reversed ordering (mismatched outperforms matched at all fractions, 0–35% ablation range). We attribute this to OLMo’s weak SI signal (mean $R^2 = 0.058$ vs. Llama’s 0.380): with noisy SI classification, the top-quartile SI heads are not reliably the most important for tokenizer-driven processing. SI-matched pruning should therefore be treated as a model-conditional tool.

J Statistical methodology

All confirmatory tests use Holm-corrected p -values within each hypothesis family. Bootstrap confidence intervals (10,000 resamples or 2,000 for Exp. 7A) for effect size estimation. Stouffer meta-analysis combines evidence across models. Exploratory analyses are clearly demarcated and non-retroactive to confirmatory labels. Preregistered hypothesis families are locked before execution with explicit confirmatory/exploratory split. The BDI, BAC, and KOA hypotheses (Exp. 8A) were preregistered before small-model execution.

Preregistration traceability artifacts. This submission includes machine-readable preregistration artifacts in the anonymous supplementary package rather than an identified external registry. Entry points are `reinforce_exp2/schemas/preregistration.schema.json`, `reinforce_exp3/schemas/preregistration.schema.json`, and `results/reinforce_exp2/A2_prediction_layer/predictions_preregistered.json`, with per-module `preregistration.json` files at the result paths listed throughout Appendix D and Appendix J. The supplementary package also provides an anonymized GitHub repository mirror containing code, execution scripts, environment files, and reproduction commands mapped to these artifact paths. Because this is an anonymous submission snapshot, we do not include an identified external registry link here; a timestamped external archival mirror will be provided at camera-ready.

K Full Path A/B Governance Matrix

The table below contains the complete `reinforce_exp2` Path A/B governance outcomes referenced in the Robustness section of the main text. Path A modules (A0–A2) are governance infrastructure and all pass. In Path B, only B2b is supported; modules B1/B2a/B3/B4 are mixed or inconclusive and are not promoted as central claims.

L Compute resources

Experiments conducted on NVIDIA A100 (40GB) and A100/H100 (80GB) GPUs. Approximate GPU-hours per experiment: Exp. 1: 24; Exp. 2: 48; Exp. 3 core: 48; Exp. 3 Phase 2: 40; Exp. 4: 16; Exp. 5: 24; Exp. 6: 12 (partial); Exp. 7A: <1; Exp. 7B: 18; Exp. 8A: 4; Exp. 8B: 24. Total: ~260 GPU-hours reported; additional compute for preliminary and failed runs not included. Exp. 8B on OLMo required a 48GB GPU (A6000) due to OOM on 16GB cards during multi-fraction evaluation.

M Completed Claim-Hardening and Attribution Controls

The following controls are completed and integrated into main-paper claim wording. Rows marked **required** were identified as validity-gap closers and are now reported with finalized outcomes.

Additional rapid hardening add-ons (E31A–E31D) are reported as a compact extension layer rather than folded into Table 21: E31A breadth consolidation over 11 models/7 families shows a $140\times$ max/min mean- R^2 spread; E31B expanded proxy coherence refresh is mixed (Pearson $r = 0.577$, $p = 0.063$; Spearman $\rho = 0.500$, $p = 0.117$); E31C functional-coupling reanalysis across E12/E29B/E30A rows is directionally positive (Pearson $r = 0.956$, one-sided grouped permutation $p = 0.0046$); and E31D Llama natural-text passkey add-on supports SI-targeted > importance-matched degradation (mean +3.64, 95% CI [2.13, 5.03], one-sided $p = 3.33 \times 10^{-5}$; 60 prompts).

These outcomes tighten claim scope and strengthen attribution posture: central results remain positive but are explicitly non-exclusive (E17) and model-conditional under strict offset-structured retrieval controls (E18), with broader probe-family transfer non-passing at model level (E28C), measurement robustness upgraded from unresolved to bounded-supported (E19), heterogeneity now variance-decomposed in proxy setup (E20), RoPE-over-NoPE separation confirmed in matched paired-seed proxy training (E21), and Result-II SI-specificity narrowed from caveated support (E24) to mixed residual enrichment after direct contrast quantification (E24b). E25 adds a multi-axis comparability panel that supports bounded cross-model ordering claims while preserving control-normalized caveats. E28A provides stronger 20-bin head-level dose-response evidence with permutation-tested p -values; E28B adds SI-vs-importance matched separation; E28D and E30B provide convergent local-bias-disambiguation support in 2/3 models; E29A adds direct kernel-transplant specificity support in all three models. These complement E24’s collective-level finding. E27 validates threshold robustness and cross-model comparability at fixed absolute R^2 threshold, with per-head disruption tracking model-level R^2 across all three models. E29B and E30A jointly provide model-conditional transfer extension (2/3 passes) with explicit non-universal boundary directionality, while E29C adds a non-promoted extra-model quick anchor that does not directionally corroborate. E31A extends heterogeneity evidence to an 11-model breadth panel, E31C adds directional functional-coupling support across existing probe suites, and E31D adds a short naturalistic SI-vs-importance intervention check. E31B keeps the coherence-gap extension explicitly mixed, so the CS lens remains framework-level.

E24 artifact root: results/reinforce_exp3/E24_non_si_matched_baseline/. E24b artifact root: results/reinforce_exp3/E24b_si_vs_non_si_contrast/. E25 artifact root: results/reinforce_exp3/E25_cross_model_comparability/. E26 artifact root: results/reinforce_exp3/E26_r2_disruption_dose_response/. E27 artifact root: results/reinforce_exp3/E27_absolute_threshold_robustness/. E28A artifact root: results/reinforce_exp3/E28a_e26_20bin_exact/. E28B artifact root: results/reinforce_exp3/E28b_importance_matched_control/. E28C artifact root: results/reinforce_exp3/E28c_probe_diversity_transfer/. E28D artifact root: results/reinforce_exp3/E28d_r2_localbias_decomposition/. E29A artifact root: results/reinforce_exp3/E29a_kernel_transplant_specificity/. E29B artifact root: results/reinforce_exp3/E29b_naturaltext_longcontext_probe/. E29C artifact root: results/reinforce_exp3/E29c_qwen_anchor_quickcheck/. E30A artifact root: results/reinforce_exp3/E30a_probe_boundary_grid/. E30B artifact root: results/reinforce_exp3/E30b_localbias_null_family/. E31A artifact root: results/reinforce_exp3/E31a_breadth_consolidation/. E31B artifact root: results/reinforce_exp3/E31b_coherence_refresh/. E31C artifact root: results/reinforce_exp3/E31c_functional_coupling_reanalysis/. E31D artifact root: results/reinforce_exp3/E31d_llama_naturalistic_importance_control/.

Residual scope note after E20/E21 completion. E20/E21 close the prior deferred-control gap for proxy attribution. Remaining limitation is scale transfer: E21 is a controlled TinyLlama-scale proxy and does not replace a full matched-scale 7–8B RoPE-vs-ALiBi/NoPE pretraining suite.

Tokenizer toy-model pretraining experiments (ongoing exploratory work). Separate exploratory toy-model tokenizer/pretraining runs are active to test when tokenizer-mediated SI routing emerges

during training. These runs are not part of the claim-hardening table above and are not used for any promoted claim in this submission.

E23: Exploratory stage-sensitive SI pretraining causal probe. E23 trains a tiny from-scratch LlamaForCausalLM (384 hidden units, 10 layers, 6 heads; vocab_size=192; synthetic arithmetic/copy tasks) with SI-consistency augmentation active in different training-stage windows. The v1 run failed due to a CE-masking bug (filler tokens dominating the gradient, discovered and fixed for v2). V2 results show bimodal outcomes: seeds either achieve 100% accuracy on unseen offsets (offset-invariant solution) or stay near 10% chance. SI augmentation at any stage (early, late, or full) increases the probability of perfect generalization relative to the no-augmentation baseline (baseline 0/2 perfect events; early 1/2; late 1/2; full 1/2). The strict stage-sensitivity hypothesis (early augmentation specifically critical; late arm stays near baseline) is not supported—the late arm improves substantially. Kernel specificity support is False in all seeds including perfect-generalization ones, indicating the model learns an offset-invariant computational strategy without raising measurable $\bar{R}_{SI}^2$. This is exploratory causal probe evidence for the training-modulates-exploitation framing. Full details in Appendix N.

N Exploratory Causal Probe: Stage-Sensitive SI Pretraining (E23)

N.1 Motivation

The capacity-versus-exploitation framing in the main text claims that RoPE provides SI capacity at initialization and training determines how strongly each model exploits it. E4 (OLMo checkpoint analysis) provides consistency evidence for architectural initiation, but cannot directly probe whether and when training modulations of SI capacity occur. E23 is an exploratory causal probe: by training a tiny RoPE model from scratch with SI-consistency augmentation active in different training-stage windows, we test whether *when* augmentation is applied influences whether the model converges to an offset-invariant computational solution. This experiment is explicitly exploratory—the synthetic setting does not generalize to 7–8B models, and no promoted claims in the main text depend on E23.

N.2 Setup

Model. Tiny LlamaForCausalLM: 384 hidden units, 10 transformer layers, 6 attention heads (GQA with num_key_value_heads=6), vocab_size=192, max_position_embeddings=256, RoPE $\theta = 10000$. Trained on synthetic sequences of length 128.

Tasks. Two synthetic tasks interleaved in each batch:

- **Arithmetic.** The model is given two key tokens (token IDs 64–95), each associated with a 1-digit value (IDs 16–25). It must predict the sum at designated answer positions. All task-relevant tokens use IDs < 128 (digits: 16–25; keys: 64–95; values: 96–127; structure: 0–10). IDs 128–191 are random fillers occupying ~92% of each sequence.
- **Copy.** Copy the first non-filler tokens from a shifted input position. Solved trivially in all arms within the first ~100 steps; provides a stable positive signal for training.

SI augmentation mechanism. Each training step draws a “base” batch and a “shifted” batch with the same task specification and content but different absolute sequence offsets. The total loss is:

$$\mathcal{L} = \mathcal{L}_{CE}(\text{base}) + \lambda \cdot \mathcal{L}_{\text{symKL}}, \quad (11)$$

where $\mathcal{L}_{CE}$ is cross-entropy masked to non-filler positions (labels < 128), and $\mathcal{L}_{\text{symKL}}$ is the Bernoulli symmetric KL divergence between answer-token probabilities at base and shifted positions ($\lambda = 0.2$). SEEN offsets used during training: {8, 16, 24, 32, 40}. UNSEEN offsets held out for evaluation: {12, 20, 28, 36, 44}.

Training arms ($n=2$ seeds each).

- **baseline:** No SI augmentation throughout training.
- **early_si_aug:** Augmentation active for training fraction ≤ 0.40 .

- 901 • **late_si_aug**: Augmentation active for training fraction > 0.60 .
- 902 • **full_si_aug**: Augmentation always active.

903 Total training budget: 40M tokens (4,883 steps at 8,192 tokens/step). Optimizer: AdamW, lr
 904 $= 3 \times 10^{-4}$, 5% linear warmup, weight decay 0.01, gradient clip 1.0.

905 **Implementation note: v1 CE-masking bug.** The original E23 run (v1) produced all-null results: all
 906 arms showed arith accuracy $\approx 10\%$ (random chance) and symKL = 0.0000 throughout. Root cause:
 907 `_forward_ce_loss` computed cross-entropy over all 128 token positions uniformly, including ~ 120
 908 random filler tokens ($\sim 92\%$ of the sequence). The arithmetic task received only $\approx 1/128$ of the
 909 total gradient budget, preventing any learning. Fix: mask CE to positions where labels < 128
 910 (FILLER_BASE), concentrating gradient on the 8–14 task-relevant tokens per sequence. Verified by
 911 CE dropping from 3.88 $\rightarrow$ 0.897 at step 100 in the repaired v2 run (v1 flatlined at 3.88 indefinitely).

912 N.3 Results

913 **Bimodal outcome distribution.** The fundamental result is a phase transition in solution type: a
 914 seed either achieves 100% accuracy on both seen and unseen offsets (the offset-invariant solution)
 915 or stays near 10% chance (the memorization/failure basin). Augmentation at any stage shifts the
 916 probability of offset-invariant convergence: baseline 0/2, early 1/2, late 1/2, full 1/2. The high
 917 seed-level variance ($\sigma \approx 0.25$ – 0.32 in completed two-seed arms) reflects the bimodal distribution;
 918 arm-level means conflate perfect-generalization and chance-level seeds.

919 **Strict stage-sensitivity hypothesis not supported.** The pre-registered gate required late augmen-
 920 tation to stay near baseline ($|\text{late} - \text{baseline}| \leq 0.02$). The late arm improved substantially (mean
 921 $\Delta = +0.269$), failing the gate. The correct characterization is that *augmentation at any stage*
 922 *increases $P(\text{offset-invariant solution})$* , not that there is a critical early-stage window.

923 **Kernel specificity.** `kernel_specificity_support = False` in all seeds, including perfect-
 924 generalization ones. Seeds that achieve 100% accuracy on unseen offsets show $\bar{R}_{\text{SI}}^2 \approx 0.39$ – 0.44 ,
 925 indistinguishable from chance-level seeds (0.40–0.43). The model learns an offset-invariant compu-
 926 tational *strategy* (correct generalization to unseen offsets) without developing a measurably more
 927 shift-invariant *attention kernel*. This is consistent with the main paper’s finding that high SI R^2 and
 928 task-level positional generalization can be dissociated.

929 N.4 Interpretation and Scope

930 E23 provides exploratory causal probe evidence that training-stage interventions can influence
 931 solution-basin convergence in a controlled synthetic setting. SI-consistency augmentation lowers the
 932 barrier to the offset-invariant solution, consistent with the main-paper framing that training modulates
 933 how the RoPE-provided SI capacity is exploited.

934 Critical limitations preclude generalization: (a) the model has 384 hidden units and 10 layers, orders
 935 of magnitude below 7–8B scale; (b) tasks are synthetic, highly structured, and involve only token-ID
 936 arithmetic, not natural language; (c) $n=2$ seeds per arm is too small for statistical confidence (the
 937 difference between 0/2 and 1/2 is not statistically distinguishable); (d) the strict stage-sensitive
 938 critical-period hypothesis is falsified. E23 is therefore exploratory motivation for future large-scale
 939 causal probing—not a promoted claim—and the results do not modify any main-text claim.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sections 1–3 state the claim boundaries explicitly (non-architecture-causal, no circuit-identity, no language-scope claim), and Results I–III align promoted claims with explicit evidence-tier definitions documented in Appendix L.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are discussed in the main text (Scope boundary, Robustness and Scope Limits, Governance alignment) and appendix governance sections, including E7 exclusion, E2 heterogeneity, E4 non-promotion, and model-conditional caveats.

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1277 be described.

Table 10: Reinforcement evidence map: sources, prereg acceptance criteria, outcomes, and claim-scope impact.

ID	Source artifact path	Preregistered criterion	Outcome	Claim-scope impact
R1	Multidomain gate summary JSON	Stable blocked or stable clean under multidomain pooled rule	Ambiguous overall; domain split is wiki ambiguous, code stable-clean, dialogue ambiguous; strict gate neither passes nor blocks	Keep Llama strict status as ambiguous; do not claim strict blocked or strict clean
R2	Dependence reanalysis claim table	Direction consistency after sequence-cluster sensitivity checks	7/8 directional checks supported; unsupported cell is OLMo boundary non-triviality	Main narrative retained with one explicit OLMo boundary caveat
R3	Core replication output directory	Core battery completion (T8, strict B, C1, C2, J) with final synthesis report	Completed; headline verdict is positive (kernel-load-bearing, strict-B clean, C1 threshold-supported)	Upgrades “pending” to three-model core-battery support with attribution caveat
R4	PE-scheme comparison JSON	Non-RoPE anchors show SI structure plus strict non-trivial boundary signal	Positive in both GPT-2 anchors (gpt2-medium and gpt2-small); 2/2 positive replications	RoPE-only confound narrowed (not eliminated)
R5	Task-grounded interaction transfer JSON	Significant interaction and sign-concordance $\geq 3/4$ per model	Interaction significant in both models, but sign transfer fails (Llama 1/4, OLMo 2/4)	Keep interaction-level claim; downgrade regime semantics to proxy-specific
R2B	OLMo boundary power reinforcement	Domain-level practical-positive support for boundary non-triviality under sequence-block inference	Stable-clean artifact status in all domains, but directional support fails in all domains (0/3)	Reinforces conservative OLMo boundary caveat language
R5B	Regime-alignment reinforcement	Significant interaction plus sign-concordance threshold in mapped regime battery	Interaction significant in both models; transfer still fails (Llama 1/2, OLMo 1/2)	Confirms interaction-level signal but keeps regime semantics proxy-specific
NEW-R12	Random-order ablation control	Threshold preference should survive non-ranked ablation orderings	Strong replication with model-conditional rates (Llama 7/10, Mistral 9/10, OLMo 10/10; 26/30 total threshold-majority)	Strengthens E2 claim against ranking-artifact criticism
NEW-R14	Kernel permutation specificity (Llama, Mistral, OLMo)	True SI-kernel perturbation should exceed offset-permuted perturbations	Supported in all three primary models (Llama 3.2774, Mistral 3.1926, OLMo 0.0427; all one-sided bootstrap $p = 2.0 \times 10^{-4}$)	Strengthens mechanistic specificity of T8 channel
NEW-R15	High-SI head identity stability	Cross-domain high-SI set overlap should exceed random expectation	Partially stable in both models (Llama 3-way Jaccard 0.594, OLMo 0.427; both above random)	Supports head-set reuse with caveat (not high-confidence stable; small code shard)

Table 11: reinforce_exp2 Path A/B evidence map: module intent, prereg/criteria gates, observed outcomes, and paper-claim impact.

Module	What was run	Prereg/criteria gate	Observed result	Claim impact	Artifact path
A0	Claim-hygiene matrix generation and status assignment	Complete claim-map with strict-only tagging	n_claims=5; status mix includes pending/proxy/mixed	Governance support with caveat	results/reinforce_exp2/A0_claim_hygiene/claim_matrix.json
A1	Evidence registry and discrepancy scan	Structural integrity checks over referenced artifacts	n_registry_rows=9; n_high_severity=0	Governance support (traceability gate passed)	results/reinforce_exp2/A1_evidence_consolidation/evidence_registry.json
A2	Prediction-layer prereg freeze	At least one directional prediction with failure interpretation	n_predictions=1 (P1_shuffle_sensitivity)	Governance support (falsifiability retained)	results/reinforce_exp2/A2_prediction_layer/predictions_preregistered.json
B1	Kernel taxonomy + prototype/enrichment gates	Criteria 1/2/6 required with depth/directional/spectral checks	Criteria 1/2/6 pass; 3/4/5 fail; Mistral shard-wise gate still has n_valid_layers=0 at 200-seq calibration; B1_supported=false	Mixed (taxonomy not promoted as main mechanistic unit)	results/reinforce_exp2/B1_kernel_taxonomy/kernel_taxonomy_summary.json
B2a	Head-level signed/absolute alignment vs null	Criteria 1–4 (variance, Holm significance, practical effect, directional)	Criterion 1 pass; 2/3/4 fail; B2a_supported=false	Mixed (no promoted cross-model directional head-level claim)	results/reinforce_exp2/B2a_head_alignment/alignment_null_tests.json
B2b	Cluster-level alignment enrichment with depth controls	Criteria 1–3 (Holm, depth retention, practical)	All criteria pass; B2b_supported=true	Supported (confirmatory reinforcement channel)	results/reinforce_exp2/B2b_cluster_alignment/summary.json
B3	Matched-triplet cluster ablation disambiguation	Cross-model mechanism verdict with Holm-controlled family	B3_inconclusive; reason insufficient_clusters; family p-values = 1.0	Mixed/inconclusive (no disambiguation promotion)	results/reinforce_exp2/B3_cluster_ablation/mechanism_disambiguation_verdict.json
B4	Confound-isolated high-vs-low SI contrasts	Criteria 1–3 (direction/statistical/practical retention)	Criteria fail overall; B4_supported=false, B4_mixed=true	Mixed (confound-isolated claim remains non-load-bearing)	results/reinforce_exp2/B4_confound_isolation/summary.json

Table 12: Exp. 4A results: all SI-routing conditions vs. baseline, by model.

Model	Comparison	Accuracy Δ	95% CI	p
Llama-3.1-8B	(a) SI-protecting vs (b) uniform	−0.011	[−0.057, +0.034]	0.668
OLMo-2-7B	(a) SI-protecting vs (b) uniform	−0.009	[−0.025, +0.006]	0.355
Llama-3.1-8B	(c) SI-amplified vs (d) baseline	−0.015	[−0.049, +0.019]	0.489
OLMo-2-7B	(c) SI-amplified vs (d) baseline	−0.003	[−0.019, +0.013]	0.742
Llama-3.1-8B	(b) uniform vs (d) baseline	−0.021	[−0.060, +0.018]	0.405
OLMo-2-7B	(b) uniform vs (d) baseline	+0.004	[−0.014, +0.022]	0.691

Table 13: Exp. 5C perturbation effects on SI attention.

Perturbation	Llama Δ attention	p	OLMo effect
Fake boundary insertion + prefix	+0.017 to +0.052	$< 10^{-21}$	Large
Character decomposition	+0.015 to +0.017	$< 10^{-34}$	Large
Random resegmentation	+0.003 to +0.005	> 0.05	Null

Table 14: Exp. 6A gradient routing results.

Model	Math Δ (pp)	p	PPL Δ	p	Interpretation
Llama-3.1-8B	−12.3	0.012	−0.535	0.003	SI disruption at scale
TinyLlama-1.1B	−0.2	0.881	−1.227	0.003	Scale-dependent

Table 15: Exp. 7A: coherence gap vs. $\bar{R}^2$ correlation results.

Metric	Value	Notes	p
Pearson r (model-level)	+0.731	$n = 6$ models	0.099
Spearman ρ (model-level)	+0.516	$n = 6$ models	0.295
Permutation (two-sided, model-level)	0.141	exact shuffles	—
Pearson r (36-row exploratory)	+0.609	pseudo-replicated rows	8.1×10^{-5}

Table 16: 8A boundary discrimination full results (small models only; 7–8B-scale pending).

Model	BDI _{SI}	$p(H1)$	ΔBDI	$p(H2)$	Cohen’s d
GPT-2-small	+0.026	1.3×10^{-15}	+0.026	6.2×10^{-15}	5.14
TinyLlama-1.1B	+0.008	4.1×10^{-12}	+0.010	1.1×10^{-13}	4.58

Table 17: 8B tokenizer-aware pruning on Llama-3.1-8B: early-ablation comparison.

Fraction ablated	si_matched math	si_mismatched math	Gap (pp)
5%	0.662	0.369	+29.3
10%	0.731	0.369	+36.2
15%	0.562	0.212	+35.0

Table 18: 8B primary matched-vs-mismatched comparison by model (AUDC summary).

Model	Matched math AUDC	Mismatched math AUDC	Mean diff	p
Llama-3.1-8B	0.261	0.198	+0.137	0.034
OLMo-2-7B	0.168	0.251	−0.160	0.015

Table 19: reinforce_exp2 Path A/B governance matrix.

Module	Intent	Decisive criterion/stat	Verdict	Paper-claim impact
A0	Claim hygiene	5 mapped claims; status counts: pending 2, proxy-specific 1, supported-with-caveat 1, mixed 1	supported with caveat	Enforces explicit status-tagged claim wording and caveat visibility
A1	Evidence consolidation	Registry rows 9; discrepancies 4; high-severity discrepancies 0	supported	Governance gate passes for claim-evidence traceability
A2	Prediction layer freeze	1 preregistered prediction (C2.1, beta-shuffle endpoint)	supported	Preserves falsifiability; reduces retrospective-only interpretation risk
B1	Kernel taxonomy	Criteria 1/2/6 pass; criteria 3/4/5 fail (Mistral shard-wise reliability gate underpowered: 200-seq calibration $\Rightarrow$ <code>n_valid_layers=0</code> ; proto-type pairing passes)	mixed	Taxonomy retained as exploratory context, not promoted as main mechanistic unit
B2a	Head-level alignment	Criterion 1 pass; criteria 2/3/4 fail (Holm: Llama absolute 1.75×10^{-16} , OLMo absolute 0.823)	mixed	No promoted cross-model directional head-level alignment claim
B2b	Cluster-level alignment	Criteria 1/2/3 pass; confirmatory clusters: Llama=3, OLMo=3, Mistral=4	supported	Confirmatory reinforcement for cluster-level alignment narrative
B3	Cluster ablation disambiguation	Inconclusive due to insufficient clusters; Holm family p -values all 1.0	inconclusive	Keep threshold-capacity wording conservative; do not promote redundancy-vs-group disambiguation
B4	Confound isolation	Criteria 1/2/3 fail; retention at model level: Llama true, OLMo false, Mistral false	mixed	Confound-isolated high-vs-low SI claim remains non-load-bearing
C1	Two-carrier-class prediction	Criteria 1–4 fail; criterion 5 (disjointness) passes	mixed/null	Non-promotional; does not demote B2b
C2	Carrier-class prediction	C2.1/C2.2 both fail strict gates; <code>C2_supported=false</code>	null	Non-promotional; strict endpoint null

Table 20: Evidence provenance map for headline claims (addresses 1B vs. 7–8B sourcing). All headline claims use 7–8B primary models; 1B models serve as scope anchors and proxy controls only.

Headline claim	Primary scale	Provenance in this paper
SI-strength heterogeneity (0.380/0.266/0.058) and kernel-subtraction load-bearing deltas (Appendix D, T8)	7–8B	Main text Results I–III; 7–8B primary models
Breadth heterogeneity extension (11-model panel; max/min mean- R^2 ratio 140 $\times$)	Mixed-scale breadth panel	Appendix L, E31A (descriptive consolidation; non-promotional for causal attribution)
NoPE anchor (early-layer $R^2 < 0.01$)	1.1B control	Main text scope anchor; detailed measurement tables in Appendix A
Checkpoint trajectory consistency (Appendix D, E4)	7B (OLMo only)	Main-text scope/consistency note; not promoted as cross-model dynamics proof
Linear rejection (0/18, 26/30, 0/6) + Gemma directional replication (0/3)	7–8B (+simplified Gemma protocol)	Appendix-first contextual ablation regularity (non-SI-unique); concise main-text scope note in robustness
Offset-repetition retrieval probe preferential degradation plus strict, broadened, semi-naturalistic, and boundary-grid transfer controls (Appendix L, E18/E28C/E29B/E30A)	7–8B	Main text Result III; strict control mixed, broader synthetic transfer non-passing (0/3), long-context transfer model-conditional (2/3), and boundary-grid transfer 2/3 with non-universal factor directionality
Naturalistic SI-vs-importance intervention validation (Llama + Mistral replication)	8B (two models)	Appendix L; cross-model replication of practical extension within Result III scope ($p=3.33 \times 10^{-5}$ both models)

Table 21: Completed hardening and attribution controls with claim impact (E17–E21, E24, E24b, E25, E26, E27, E28A, E28B, E28C, E28D, E29A, E29B, E29C, E30A, E30B).

ID	Priority	Why required	Finalized outcome	Claim impact
E17	Required	T8/NEW-R14 do not isolate SI-specific disruption from generic large-amplitude perturbation.	<code>not_supported</code> : in all 3 models, true subtraction exceeds permuted control but is weaker than norm-matched random perturbation.	Result I retained as load-bearing under intervention; SI-specific exclusivity downgraded.
E18	Required	E12 directional result required stricter format-confound adjudication.	<code>mixed / model_conditional</code> : SI-selective degradation survives full control battery in 1/3 primary models.	Result III wording remains directional and model-conditional under strict controls.
E28C	Hardening	Result-III scope required testing beyond a single maximally aligned 9-token probe format.	<code>not_supported</code> : broader 7-family probe-diversity transfer does not pass model-level gates (0/3 model passes; Llama 4/7, Mistral 4/7, OLMo 2/7 LP family passes; pass threshold 6/7).	Result III narrowed to controlled offset-repetition probe sensitivity with non-passing broader-format transfer.
E19	Required	Track-A SI score may be influenced by offset support and corpus composition.	<code>supported / robust_cross_model</code> : bounded deltas under offset reweighting, equal-count proxy, and corpus-transfer scoring with preserved cross-model contrast.	Measurement caveat narrowed from unresolved to bounded-supported.
E20	Required	Needed to separate cross-model heterogeneity from training-run seed variance under matched setup.	<code>model_family_dominant</code> : between-model SI variance 0.04074 vs. within-model seed variance 0.000223 (ratio 182.73; 9/9 seeds complete).	Heterogeneity claim upgraded from descriptive-only to variance-decomposed in tested proxy setup; still non-causal across confounded family differences.
E21	Required	Needed to move beyond single NoPE anchor and test controlled RoPE-vs-NoPE SI separation.	<code>rope_gt_nope_proxy</code> : paired TinyLlama $\Delta R^2 = 0.4665$ (95% CI [0.4553, 0.4872]); RoPE 0.5511 vs. NoPE 0.0846 (6/6 seeds complete).	Capacity framing strengthened with matched proxy evidence; matched-scale 7–8B PE-family generalization remains open.
E24	Required	Result-II SI-specific organization required matched non-SI cumulative-ablation baseline.	<code>supported_with_caveat</code> : SI-ranked cells 0/9 linear votes; matched non-SI cells 1/9 linear votes (8/9 non-linear).	Result II reframed as strong SI-ranked linear rejection with limited SI-specificity evidence; generic nonlinearity remains plausible.
E24b	Required	E24 left residual SI-enrichment magnitude unresolved; needed direct SI-vs-non-SI contrast quantification.	<code>mixed</code> : 5/9 SI-vs-non-SI contrast cells favor stronger SI nonlinearity, but cross-cell uncertainty spans zero (delta-BIC CI includes 0).	Result II narrowed further to ranked-ablation non-linearity; SI-specific enrichment treated as mixed/non-decisive.
E25	Supporting	Cross-model Result-I comparability required explicit baseline-normalized and control-normalized sensitivity panel.	<code>supported_with_caveat</code> : raw-nat and percent rankings align, but control-normalized ranking is mixed.	Main text now reports cross-model contrasts on multiple axes and keeps bounded comparability caveat explicit.
E26	Hardening	Result-I and E24 left head-level $R^2 \rightarrow$ disruption specificity untested under the original 5-bin design.	<code>supported</code> : original quintile dose-response is positive in all three models; final interpretation now relies on the stronger E28A 20-bin hardening and E28B/E28D controls.	Serves as predecessor evidence; main-text numeric reporting promoted from E28A/B/D.
E27	Hardening	Weakness-1 (within-model threshold not principled): does the rank-based SI-disruption	<code>supported</code> : all 18 threshold $\times$ model conditions show true > permuted (frac true > perm	Result-I threshold choice validated; cross-model per-

Table 22: E23 v2 seed-level evaluation results available at submission cutoff. arith_seen/unseen: accuracy on seen/unseen offsets. joint_unseen: geometric mean of arith_unseen and copy_unseen (copy_unseen=1.00 in all seeds). $\bar{R}_{\text{SI}}^2$: mean per-head SI score across all heads. kernel_spec: whether the kernel_specificity gate passes.

Arm	Seed	arith_seen	arith_unseen	joint_unseen	$\bar{R}_{\text{SI}}^2$	kernel_spec
baseline	0	0.100	0.082	0.541	0.425	×
baseline	1	0.083	0.128	0.564	0.431	×
early_si_aug	0	1.000	1.000	1.000	0.439	×
early_si_aug	1	0.110	0.108	0.554	0.416	×
late_si_aug	0	0.247	0.285	0.643	0.360	×
late_si_aug	1	1.000	1.000	1.000	0.402	×
full_si_aug	0	1.000	1.000	1.000	0.395	×
full_si_aug	1	0.082	0.102	0.551	0.357	×

Table 23: E23 v2 arm means at submission cutoff. Std is across available seeds.

Arm	joint_unseen ($\mu \pm \sigma$)	arith_unseen mean	$\bar{R}_{\text{SI}}^2$	perfect-gen events
baseline	0.553 ± 0.017	0.105	0.428	0/2
early_si_aug	0.777 ± 0.315	0.554	0.427	1/2
late_si_aug	0.821 ± 0.253	0.642	0.381	1/2
full_si_aug	0.775 ± 0.318	0.551	0.376	1/2